Adjustable Step-Down Converter Based on the LM2596

Electronic Engineering Technical Documentation

Project Logbook

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Abstract

This document describes the design, KiCad schematic/PCB layout, and COMSOL thermal validation of an adjustable step-down (buck) power supply built around the LM2596S-ADJ switching regulator. It follows the same project series as the LM317 linear adjustable supply (`FuenteLM317_9V`): a quick, practical exercise meant primarily to build real PCB layout experience (placement, routing priorities, a ground-plane strategy, via stitching under a thermal pad) and to validate the result thermally, before moving on to multiphase VRM work. Documented as an honest engineering logbook: every result is marked as **calculated**, **simulated**, or **pending** – never filled in with invented placeholder data. LTspice simulation of the closed loop was deliberately skipped for this project (see Section 2): the topology and component values are taken directly from the manufacturer's own validated application circuit, so re-deriving them in SPICE would have added time without adding information. **Key result**: at the demonstration operating point (12 V → 2 V, 0.5 A), COMSOL gives 0.444 W total estimated loss, with the LM2596's junction at 44.3 °C and the catch diode (the hottest point on the board) at 52.5 °C – both comfortably under their rated limits. A separate worst-case sweep (35 V→1.23 V, the extreme low-duty-cycle end of the adjustable range) found that the catch diode, not the regulator IC, is the component that actually limits the safe operating envelope, exceeding its rated junction temperature above ≈1.5–2 A in the conservative (no-radiation) case. This is documented as a known limitation with a proposed, not-yet-implemented fix.

Contents

1 Project Objective **3**

2 Reference Design **3**
2.1 Why no closed-loop SPICE simulation . . . 3
2.2 Topology and component values (adopted from the datasheet) . . . 3

3 KiCad Schematic and PCB **4**
3.1 Footprint selection . . . 4
3.2 Capacitor sizing . . . 4
3.3 Layout priorities applied . . . 4
3.4 Ground strategy . . . 4
3.5 Via stitching under U1's thermal pad . . . 5
3.6 ERC / DRC . . . 5

4 Thermal Simulation (COMSOL) 5
4.1 Methodology . . . 5
4.2 Model setup in detail . . . 5
4.3 Demonstration operating point: 12 V→2 V at 0.5 A . . . 7
4.4 Worst-case sweep: 35 V→1.23 V (minimum duty cycle) . . . 8
4.4.1 Variants at 3 A (worst case) . . . 9
4.5 Root cause and proposed improvement . . . 9
4.6 Modeling and debugging log . . . 10
4.7 Refinements attempted but not adopted . . . 10

5 Documented Design Limitation 11

6 Conclusions 12

7 License 12

1 Project Objective

Design, lay out, and thermally validate an adjustable step-down converter using a real integrated buck regulator (LM2596S-ADJ), with two explicit goals: (1) build hands-on KiCad PCB layout experience – component placement for a switching converter, trace-width sizing, a two-layer ground-plane strategy, and via stitching under a thermal pad – as a deliberately fast, low-complexity exercise before the next multiphase VRM project; and (2) validate the design thermally in COMSOL, in the same style as the LM317 project, so the two can be compared directly (Section 6).

2 Reference Design

2.1 Why no closed-loop SPICE simulation

Unlike the LM317 and VRM projects in this series, this project does **not** include an LTspice closed-loop simulation of the LM2596. The topology and component values below are taken directly from the manufacturer's own datasheet application circuit (Texas Instruments LM2596 datasheet, Figure 30: "1.2 to 35 V Adjustable 3.0 A Power Supply with Low Output Ripple"), which is itself a validated, published reference design. Re-deriving a SPICE model for an extremely standard, widely-used topology would have repeated work the manufacturer already did, without adding new information – so the LTspice step was intentionally skipped to prioritize time on the PCB layout and the thermal validation, which *are* specific to this board.

2.2 Topology and component values (adopted from the datasheet)

U1	LM2596S-ADJ (adjustable step-down regulator, TO-263-5 / PowerPAK-sty
C1 (input)	100 µF, rated ≥50 V (covers the 40 V max input)
D1 (catch diode)	SS34 Schottky, 3 A/40 V (SMD substitute for the datasheet's 1N5822, sam
L1	33 µH
C2 (output)	220 µF
RV1	50 kΩ trimmer potentiometer (feedback, adjustable)
R1	1.21 kΩ (feedback, fixed)
L2, C3 (optional output ripple filter)	20 µH, 100 µF

Output voltage range (datasheet formula, $V_{ref} \approx 1.23\,\text{V}$):

$$V_{out} = V_{ref}\left(1 + \frac{R_{RV1}}{R_1}\right) \tag{1}$$

With $R_1 = 1.21\,\text{k}\Omega$ and $RV1$ swept 0 to 50 kΩ: V_{out} spans 1.23 V (RV1 at 0 Ω) up to the catalog figure's stated 35 V at a partial (not full-scale) RV1 setting – the datasheet's own "1.2 to 35 V" range does not correspond to the pot's full 0–50 kΩ travel at these exact R1/RV1 values (full travel would mathematically reach ≈52 V), so the usable 35 V ceiling is set by the pot position, not derived from scratch here – adopted as given in the reference.

The optional output ripple filter (L2/C3) sits *after* the point where feedback is sensed – the regulator regulates the pre-filter node, and L2/C3 only clean up additional high-frequency ripple on top of that already-regulated voltage, so the extra filter does not fight the control loop.

3 KiCad Schematic and PCB

3.1 Footprint selection

Part	Footprint	Note
U1 (LM2596S-ADJ)	`Package_TO_SOT_SMD:TO-263-5_TabPin3`	Stock KiCad symbol/footprint pair
D1 (SS34)	`Diode_SMD:D_SMC`	SMD, same 3A/40V rating as the datasheet's 1N5822
C1, C2, C3	`Capacitor_THT:CP_Radial_D8.0mm_P3.50mm` / `D10.0mm_P5.00mm`	Sized per value/voltage, Section 3.2
RV1 (50k trimmer)	`Potentiometer_THT:Potentiometer_Bourns_3266Y_Vertical`	Real, common part
R1	`Resistor_SMD:R_0805_2012Metric`	Consistency with the rest of the SMD parts
L1, L2	`Inductor_SMD:L_12x12mm_H4.5mm`	SMD shielded power inductor (no 3D model in the KiCad library, see Section 4.2)
TP1–TP4 (I/O)	`TestPoint:TestPoint_THTPad_3.0x3.0mm_Drill1.5mm`	Bare solder pads, not a connector housing

3.2 Capacitor sizing

Radial electrolytic package sizes were chosen from the actual voltage rating each capacitor needs, not a single generic size for all three:

Cap	Value	Voltage needed	Typical minimum can size
C1	100 µF	∼50 V	∅8x11.5 mm
C2	220 µF	∼35 V	∅10x12.5 mm
C3	100 µF	∼35 V	∅8x11.5 mm

PENDING

These are typical catalog sizes for the stated value/voltage, not a specific vendor part number verified against a real datasheet (unlike the LM317 project's heatsink and capacitor choices). Low risk for a layout-practice board, but should be confirmed if this were ever built.

3.3 Layout priorities applied

Following the TI application note's own layout guidance ("locate the programming resistors near the feedback pin using short leads" and "keep feedback wiring away from inductor flux"):

- RV1/R1 (the feedback divider) placed with short, direct traces to the FB pin, and physically separated from L1's body – not adjacent to it the way some low-cost commercial LM2596 modules do (a deliberate choice discussed and compared against a real commercial module photo during layout).
- High-current power traces (V_in, the switch node, L1–C2–L2–C3, V_out) sized to ≈1.3 mm (conservative, 10°C-rise IPC-2221 estimate for 3 A on 1 oz copper) – not "as narrow as possible", which was an incorrect instinct corrected during layout. Control/feedback traces (FB, ON/OFF) kept narrow, since they carry negligible current and narrower traces reduce coupling area.
- Short, direct traces between U1's output, D1, and L1 (the switching loop) instead of a copper pour – at 3 A and with a simple linear (non-branching) node topology, a pour was judged unnecessary (see Section 6 discussion, contrasted with the VRM project's switch node, where a pour was the right call at 10–20 A).

3.4 Ground strategy

Two-layer board (unlike the VRM project's four layers – a single ground domain and 3 A do not justify the extra layers and SGND/PGND split that the VRM required). Top layer: components

and all signal/power routing. Bottom layer: a solid ground pour, connected up via stitching vias placed next to each ground pad (not via-in-pad, to avoid a solder-wicking risk relevant on a board meant to represent real assembly practice) – through-hole parts' own leads double as their via where the lead already reaches the bottom layer.

3.5 Via stitching under U1's thermal pad

U1's exposed GND/thermal pad (pin 3 of the TO-263-5) is stitched to the bottom-layer ground plane with **9 vias** in an even 3×3 grid, standard via size, well clear of the pad edge. This is more than the 4–6 vias that would be strictly necessary at 3 A for this package, but causes no harm; it was kept rather than reworked.

3.6 ERC / DRC

Schematic ERC and PCB DRC both clean (confirmed in KiCad, no outstanding violations). Not intended for fabrication.

4 Thermal Simulation (COMSOL)

4.1 Methodology

Heat Transfer in Solids, natural convection, 25 °C quiescent ambient air, no heatsink. The PCB copper (top/bottom masks) and component geometry were exported from the KiCad design (`fem/geometria/LM259.ste` `Arreglo.step`, plus the copper mask text files) and imported into COMSOL. Heat sources (U1, D1, L1, L2) are driven by **analytically estimated losses**, not measured or chip-vendor-simulated values:

PENDING

The loss values feeding every thermal result below are datasheet-derived estimates (V_{sat} and V_f taken at their 3 A values and held *constant* at every current – conservative at lower currents, where the real drops are smaller – plus an assumed 100 ns switching transition time; exact formulas in Section 4.2) – not measured hardware data, and not output from a manufacturer SPICE model (none is available for the LM2596; see the project's earlier LTspice-vs-COMSOL tooling discussion). The COMSOL *thermal* solve itself is a genuine FEM result for whatever loss number it is given; the uncertainty is entirely in that input loss estimate, which is conservative by construction (ignores radiative cooling, discussed next).

4.2 Model setup in detail

Software and workflow. COMSOL Multiphysics 6.4, driven entirely from Python (`MPh` library) so every model can be rebuilt from scratch: scripts in `fem/codigo/`. The final model file is `fem/BUCK_LM2596_termico_STEP.mph` (worst-case sweep and variants) and `fem/BUCK_LM2596_termico_12V_2V_05A.mph` (demonstration point). The hand-prepared geometry file `fem/BUCK_LM2596_termico.mph` is never modified by the scripts; everything is written to new files.

Geometry. The board was exported from KiCad as STEP and imported into COMSOL by hand (*Form Union*, so touching parts share faces and no manual contact pairs are needed – unlike the LM317 project). From that import the scripts then:

- **remove** the STEP's copper solids (48 track/zone objects, 106 pads, 32 via barrels/cores, and the separate 35 µm B.Cu plane object), which could not be meshed within the available memory (Section 4.6); copper is modeled as a thin layer instead (below);
- **add solder blocks** (SAC305) under U1's tab, under D1's two terminals and under U1's five pin feet: KiCad's 3D models start at $z = 1.595$ mm while the board top is at 1.51 mm, so without them the components float ≈85 µm above the board and are thermally disconnected (the LM317 "disconnected assembly" problem in a new form);
- **add 16 vias** as solid cylinders (⌀0.3 mm) at the exact KiCad via positions, with an equivalent conductivity $k_{via} = 122\,\mathrm{W\,m^{-1}\,K^{-1}}$ (25 µm copper barrel in a 0.3 mm drill);
- **add L1/L2 pads** as 40 µm copper blocks. The inductor *bodies* are not modeled: KiCad has no 3D model for `L_12x12mm_H4.5mm`, they are not in the main heat path, and their losses enter through their pads (conservative: the bodies' extra cooling area is missing). RV1 is also omitted (negligible current, no dissipation).

Copper as a thin layer with the real layout. The F.Cu and B.Cu layers were exported from KiCad as SVG (`kicad-cli pcb export svg`, board area only), rasterized at 0.1 mm and stored as copper-fraction maps (`fem/geometria/mascara_cobre_F.txt`, `_B.txt`; F.Cu 19.5 % coverage = pads and tracks, B.Cu 93.3 % = ground pour). Each map drives a COMSOL *Thin Layer* (conductive, 35 µm) on the board's top/bottom face with $k = k_{FR4} + (k_{Cu} - k_{FR4}) \cdot m(x, y)$, so every track, pad and pour clearance conducts exactly where it is on the real board, without meshing 35 µm solids. Note: the F.Cu ground zone exists in the KiCad file but is *unfilled*, so it is (correctly) not in the map.

Materials.

Part	k [W m^{-1} K^{-1}]	ρ [kg m^{-3}]	C_p [J kg^{-1} K^{-1}]	Source
FR4 board	0.3	1900	1369	typical
Copper (tab, pins, layers, L pads)	400	8960	385	standard
Vias (equivalent)	122 (0.3 off)	8960	385	barrel geometry
Solder SAC305	58	7400	230	typical
EMC (U1 and D1 bodies)	0.7	1900	900	as LM317
Filler (C1–C3, R1)	0.2	1050	1400	as LM317

Only k matters for the stationary results reported here.

Heat sources (all in *Heat rate* mode, i.e. total watts). U1 on the internal face between its copper tab and the plastic body (where the die sits, as in the LM317 model); D1 on the faces where its terminals meet the solder; L1/L2 on the top faces of their pads. Losses ($V_{sat} = 1.16$ V, $I_q = 5$ mA, $f_{sw} = 150$ kHz from the TI datasheet; $V_f = 0.5$ V from the SS34 datasheet maximum; $t_{sw} = 100$ ns, $DCR_1 = 40$ mΩ, $DCR_2 = 25$ mΩ **assumed**):

$$D = \frac{V_{out} + V_f}{V_{in} - V_{sat} + V_f}, \qquad P_{U1} = V_{sat} I_{out} D + \tfrac{1}{2} V_{in} I_{out} t_{sw} f_{sw} + V_{in} I_q,$$
$$P_{D1} = V_f I_{out} (1 - D), \qquad P_{L1} = I_{out}^2 DCR_1, \qquad P_{L2} = I_{out}^2 DCR_2.$$

Convection. Same criterion as the LM317 project (one correlation per face orientation, no air domain), but written as explicit expressions because COMSOL's built-in natural-convection correlations returned NaN in this model (Section 4.6). A single *Heat Flux* on every exterior face picks

the correlation from the face normal: horizontal plate facing up ($Nu = 0.54\,Ra^{1/4}$, or $0.15\,Ra^{1/3}$ above 10^7) and facing down ($Nu = 0.27\,Ra^{1/4}$) with $L = 10\,\text{mm}$ (board area/perimeter), vertical wall (Churchill–Chu) with $L = 5\,\text{mm}$; air properties at the film temperature. Radiation (surface-to-ambient, $\varepsilon = 0.9$) is available as a switch and *off* in all design-relevant results.

Mesh and studies. Free tetrahedra with the geometry stretched 10× in z while meshing (`zscale`), so thin parts get flat elements instead of millions of tiny ones: 110,446 elements. Studies: (1) single operating point, (2) I_{out} sweep 0.5 A to 3 A, (3) variants (vias on/off, radiation on/off, L losses on/off). Solved with a 8 GB memory cap (`systemd-run -p MemoryMax`) after an earlier unbounded run crashed the machine.

4.3 Demonstration operating point: 12 V→2 V at 0.5 A

This is the same load condition used for the LM317 comparison in the companion project, so the two can be compared on equal footing.

Duty cycle (real, incl. V_f/V_{sat} drops)	0.2205
P_{U1}	0.2329 W
P_{D1}	0.1949 W
P_{L1}	0.010 W
P_{L2}	0.006,25 W
P_{tot}	0.444 W
Output power	1.0 W
Estimated efficiency	69.25 %
U1 junction (tab-body face)	44.29 °C
D1 max (hottest point on the board)	52.49 °C

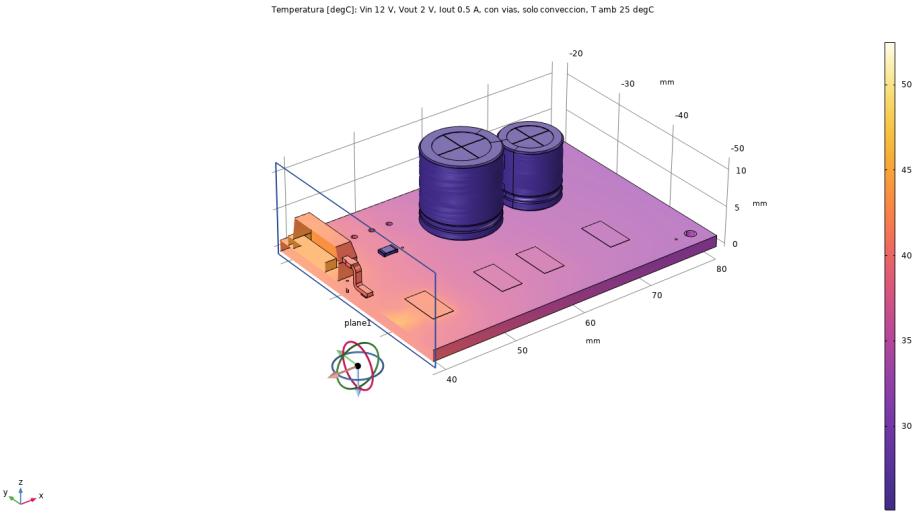


Figure 1: Temperature map, demonstration point (12 V→2 V, 0.5 A, with vias, convection only). Scale 25–55 °C.

Both results sit well under their rated limits (125 °C). Note the estimated efficiency here (69 %) is *not* representative of a buck converter's typical best-case figure (85–95 %, quoted as general knowl-

edge, not as a claim about this specific operating point) – a 2 V output next to a Schottky diode's ≈0.5 V forward drop is an inherently unfavorable ratio for a non-synchronous topology; see the worst-case sweep below for where this effect is most pronounced.

4.4 Worst-case sweep: 35 V→1.23 V (minimum duty cycle)

A second analysis deliberately targets the adjustable range's extreme low-duty-cycle end – maximum input voltage, minimum output voltage – where the catch diode conducts for the largest fraction of the switching period (duty cycle $D = (V_{out} + V_f)/(V_{in} - V_{sat} + V_f) \approx 5.0\,\%$ including the diode and switch drops, so D1 conducts ≈95 % of the time).

I_{out}	P_{tot}	U1 junction	D1 max
0.5 A	0.59 W	50.4 °C	58.9 °C
1.0 A	1.04 W	63.8 °C	85.4 °C
1.5 A	1.52 W	77.0 °C	111.4 °C
2.0 A	2.03 W	90.0 °C	136.9 °C
2.5 A	2.57 W	103.1 °C	162.5 °C
3.0 A	3.15 W	116.2 °C	187.9 °C

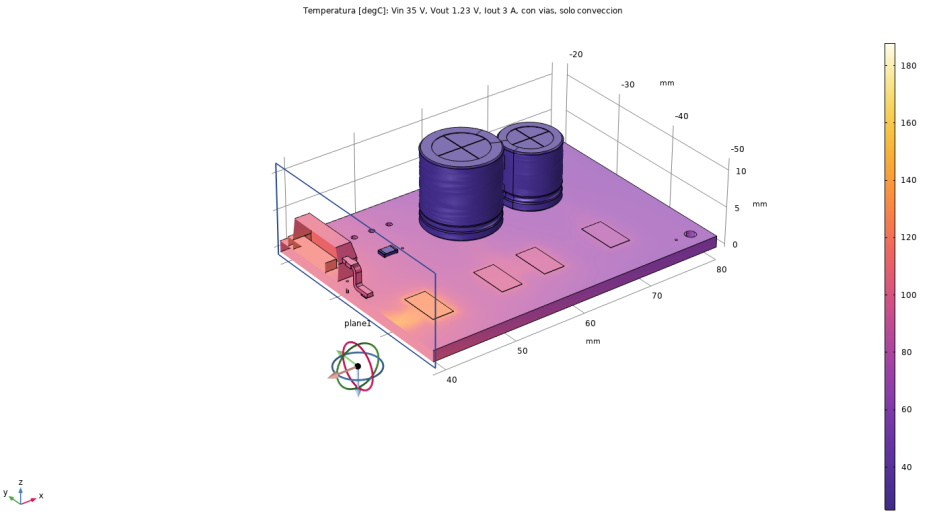


Figure 2: Temperature map, worst case at 3 A (35 V→1.23 V, with vias, convection only). Scale 25–190 °C; the hot spot is D1's anode terminal (Section 4.5).

Key finding: the catch diode D1, not the LM2596 itself, is the component that limits the safe operating envelope. The SS34's rated junction temperature is 125–150 °C depending on the manufacturer; in this conservative (no-radiation) case, D1 exceeds that limit somewhere between 1.5 and 2 A at this voltage extreme, while U1 itself stays under 125 °C through the full 3 A sweep (with little margin left at the top end).

4.4.1 Variants at 3 A (worst case)

Variant	U1 junction	D1 max
With vias (base case)	116.2 °C	187.9 °C
Without vias	123.8 °C (+8)	205.2 °C (+17)
With vias + radiation ($\varepsilon = 0.9$)	89.3 °C	155.8 °C
Without vias + radiation	96.7 °C	170.6 °C
With vias, no L1/L2 losses	108.5 °C	180.6 °C

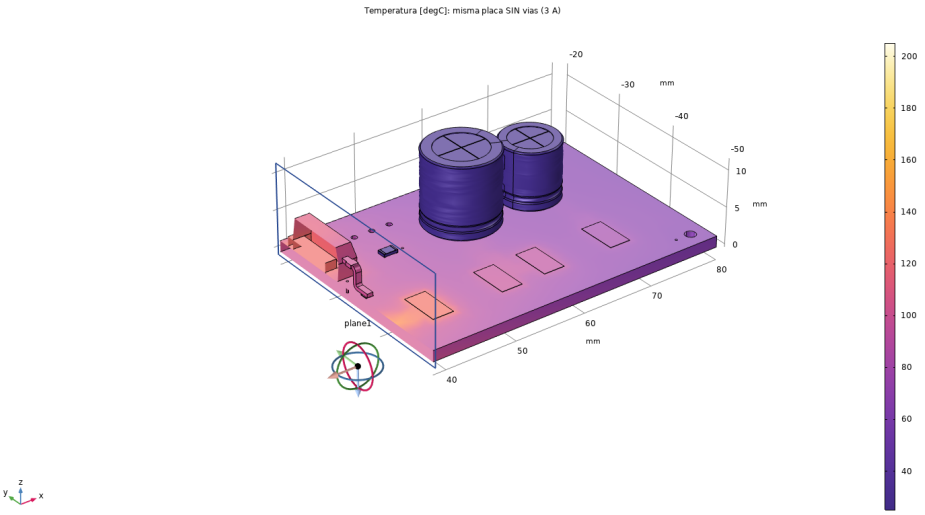


Figure 3: Same worst-case condition, **without** via stitching under U1's thermal pad. Scale 25–205 °C. Compare against Figure 2 (with vias) – the hot spot stays at D1's anode either way, confirming the vias help U1 and D1 globally but do not address D1's locally isolated terminal (Section 4.5).

Via stitching under U1's thermal pad does help (8 °C on U1, 17 °C on D1 at the worst point), but is not sufficient on its own to bring D1 under its limit at high current. Radiative cooling is large (27–32 °C) but was not used as design margin, since it is not guaranteed in an enclosure – the convection-only case is treated as the design-relevant one throughout.

4.5 Root cause and proposed improvement

The 187.9 °C peak on D1 is localized at its anode terminal – the point where the heat source enters the model. That terminal lands on a ground pad that is isolated from the main ground pour, connected to GND by only a single via. The board's center-of-die temperature a few millimeters away is much lower (≈122 °C under D1's body center at the same worst-case point), consistent with that terminal specifically being under-connected thermally, not the whole diode footprint.

PENDING

Proposed improvement, not implemented in this revision: extend the F.Cu ground pour around D1's anode pad and add more stitching vias there, matching what was already done for U1's thermal pad. Expected to meaningfully reduce the 187.9 °C figure, based on

the 17 °C improvement vias already produced elsewhere on the same net – not re-simulated to confirm the exact new number, since reworking the layout for this one finding was judged not worth it for a layout-practice project at this stage (see discussion in Section 6).

4.6 Modeling and debugging log

In order, every problem found while building the model and what was done about it. Several early numbers were wrong and are recorded here explicitly so they are never reused.

1. **First model built from primitives** (blocks/cylinders with dimensions read from the KiCad STEP via FreeCAD). Superseded once the full STEP import was done by hand in COMSOL, which is the geometry used for all results.
2. **Built-in natural convection returned NaN.** Isolated by switching boundary features on one at a time on a model that solved with a fixed temperature: the *Horizontal plate, upside* correlation (and, once replaced, the downside one) produced NaN even with a raised initial temperature or different characteristic lengths. Fix: the same correlations written explicitly (Section 4.2), with a floor on ΔT and an initial temperature 20 K above ambient (the $\Delta T^{1/4}$ term has an infinite derivative at $\Delta T = 0$).
3. **Empty heat-source selection.** A box selection meant to catch U1's tab/body face caught nothing ("inside" condition with a face edge exactly on the box limit), so 1.1 W was spread over zero area. Fix: "intersects" selections in the middle of each part, and every source selection is now checked to be non-empty before solving. A second version caught too many faces (side walls of solder and pads); fixed with one "inside" box per terminal/pad.
4. **Solder gap.** Discovered by measuring every one of the 271 imported domains: components start at $z = 1.595$ mm and copper ends at 1.55 mm. Fixed with the solder blocks above.
5. **STEP copper could not be meshed.** With the imported 35–40 µm copper laminae and 25 µm via barrels, the mesher failed with "intersecting edge elements" at a different tiny feature each time (test-point pads, via annular rings), for four different size/scaling combinations. A finer mesh used 11.5 GB of RAM and was killed by the operating system (out of memory), taking the desktop session with it. Fix: copper as a thin layer with the real layout (Section 4.2), which meshes in seconds.
6. **Copper map unit bug – first results invalid.** The interpolation function ignored its "mm" argument unit and read coordinates in metres, so the copper map evaluated to 0 everywhere and the board behaved as bare FR4. Detected because the bottom ground plane ranged 25 °C to 253 °C (a solid copper plane must be nearly isothermal). The results from that run (U1 154.6 °C, D1 319.6 °C at 35 V/3 A) are **invalid**. Fix: dimensionless arguments in mm; verified that the map is 1 in the middle of the ground plane and that the plane is now nearly isothermal. All numbers in this report come from the corrected model.
7. **Sweep results read incorrectly**, then fixed: a sweep of a non-geometric parameter is stored by COMSOL as inner solutions of one dataset, not as outer ones.

4.7 Refinements attempted but not adopted

Two refinements were attempted for the demonstration point (12 V→2 V, 0.5 A) to cross-check the adopted model. **Neither produced a valid result**; they are documented so the attempt is not repeated blindly.

Solid copper from the STEP (`fem/BUCK_LM2596_termico_STEP_cobre.mph`). Same model, but keeping the STEP's tracks, pads and ground plane as solids. An automatic loop refined the mesh only around each location where the mesher failed (`fem/codigo/malla_automatica.py`). After nine local refinements, each new attempt taking longer (1→17 min), the job exceeded a 14 GB memory cap and was killed. **Conclusion:** not feasible on this machine (23 GB RAM); the thin-layer model, which uses the same real copper layout, is the adopted one.

Conjugate heat transfer with simulated air (`fem/BUCK_LM2596_termico_12V_2V_05A_aire.mph`). Instead of convection correlations, an air box around the board (finally 100×70×75 mm, open boundaries at 25 °C) with laminar flow, gravity and the *Non-Isothermal Flow* coupling (air as an ideal gas, so its density change drives the buoyancy). Copper as thin layer; THT holes filled with solder and capacitor leads cut at the board surface (both needed for the air to mesh). Attempts:

Attempt	Outcome
Stationary, 120 mm box, power ramp 10/30/60/100 %	Did not converge (Newton iteration limit) after 1.5 h.
Stationary, 100 mm box	**Invalid**: a classification threshold ("wider than 100 mm") missed a box exactly 100 mm wide, so the air was assigned FR4 and the flow had no domain. Fixed (threshold 90 mm).
Stationary with pseudo time stepping, 200 iterations	Did not converge after 78 min.
Transient to steady state (solid heat capacity scaled by 0.01 to reach equilibrium faster; equilibrium does not depend on it)	Ran to the end, but **not physical**: U1 temperature rose linearly to 171 °C at 300 s and was still rising (4.5 K per 10 s), consistent with the heat accumulating in the box instead of being carried out by the air.

The open boundaries and the gravity node were checked and are correctly set; the cause is not yet identified (the next thing to check is whether the fluid heat-transfer feature is actually using the computed air velocity).

PENDING

Conjugate (simulated-air) cross-check: not completed. No number from the air model is valid and none is used anywhere in this report. The adopted, validated result for the demonstration point remains the correlation-based one: 44.3 °C on U1 and 52.5 °C on D1.

5 Documented Design Limitation

This board's full "1.2 to 35 V, up to 3 A" adjustable range, as inherited from the datasheet reference design, is not uniformly safe. At the low-output-voltage extreme of that range, D1 exceeds its rated junction temperature above ≈1.5–2 A (Section 4), even though the demonstration point used elsewhere in this project (12 V→2 V, 0.5 A) is comfortably safe. A real safe-operating-area boundary, as a function of both V_{in}/V_{out} ratio and load current, has not been fully mapped – only the two points above and the 3A worst-case sweep. This is a genuine finding from the analysis, not a defect hidden from this report.

6 Conclusions

1. The LM2596S-ADJ reference design (TI datasheet Figure 30) was adopted directly rather than re-derived in SPICE, since re-simulating an already-validated manufacturer reference circuit would not have added information – time was spent instead on the PCB layout and thermal validation, which are specific to this board.

2. At the demonstration condition (12 V→2 V, 0.5 A), COMSOL gives 0.444 W total loss, 44.3 °C on the LM2596's junction, and 52.5 °C peak on the catch diode – both well under their rated limits, and a large, genuine improvement over the companion LM317 linear-regulator project's result at the same load (5.37 W, ≈144 °C with a heatsink, per that project's own COMSOL validation).

3. **Key limitation found:** at the adjustable range's low-output-voltage extreme (35 V→1.23 V), the catch diode D1 – not the regulator IC – is the component that actually limits safe operation, exceeding its rated temperature above ≈1.5–2 A in the conservative (no-radiation) case. Root-caused to a single isolated ground via at D1's anode pad (Section 4.5); a layout fix is proposed but not implemented in this revision.

4. Via stitching under a thermal pad is confirmed to help (8–17 °C at the 3 A worst case) but is not, by itself, sufficient to fix the D1 finding above – more copper specifically at D1's terminal is what the analysis points to, not just more vias generally.

5. All reported loss figures are datasheet-derived analytical estimates (Section 4), not measured or vendor-simulated data; the thermal FEM solve itself is a genuine result conditioned on those inputs, and is conservative in that it omits radiative cooling.

6. The PCB copper is modeled with its real layout as a thin layer driven by maps exported from KiCad. Modeling the STEP's copper as solids was attempted and is not feasible within this machine's memory (Section 4.7).

7. A conjugate heat-transfer cross-check with simulated air was attempted for the demonstration point and did not produce a valid result (Section 4.7); it remains open. An early run with a unit bug in the copper map produced invalid numbers, which are recorded in Section 4.6 and not used.

7 License

MIT – see `LICENSE` in the repository root.